

$$\Rightarrow (VD^{-1}U^T)A = I \text{ (By the defn. of inverse)}$$

By Prob 25 from Exercises 2.3, A^{-1} is invertible and $A^{-1} = VD^{-1}U^T$

$$\text{We know that } (E_n E_{n-1} \dots E_2 E_1) D = I$$

$$((E_n E_{n-1} \dots E_2 E_1) D) D^{-1} = I D^{-1} \Rightarrow (E_n E_{n-1} \dots E_2 E_1) (D D^{-1}) = I D^{-1}$$

$$\Rightarrow (E_n E_{n-1} \dots E_2 E_1) I = I D^{-1} \text{ (Defn. of inverse)} \quad (\text{Theorem 2(c)})$$

$$\Rightarrow D^{-1} = E_n E_{n-1} \dots E_2 E_1 \text{ (Theorem 2(e))}$$

$$\therefore A^{-1} = V (E_n E_{n-1} \dots E_2 E_1) U^T \quad \square$$

24) (QR Factorization) Suppose $A = QR$, where Q and R are $n \times n$, R is invertible and upper triangular, and Q has the property that $Q^T Q = I$. Show that for each b in $\mathbb{R}^n$, the equation $Ax = b$ has a unique solution. What computations with Q and R will produce the solution?

Solution: By Defn. of inverse, there exists a unique $n \times n$ matrix R^{-1} s.t.

$$RR^{-1} = R^{-1}R = I.$$

By Prob 25 from Exercise 2.3, Q is invertible and $Q^{-1} = Q^T$.

$$A = QR \Rightarrow Q^{-1}A = Q^{-1}(QR) \Rightarrow Q^{-1}A = (Q^{-1}Q)R \text{ (Theorem 2(a))}$$

$$\Rightarrow Q^{-1}A = IR \text{ (Defn. of inverse)} \Rightarrow Q^{-1}A = R \text{ (Theorem 2(e))}$$

$$\Rightarrow R^{-1}(Q^{-1}A) = R^{-1}R \Rightarrow R^{-1}(Q^{-1}A) = I \text{ (Defn. of inverse)} \Rightarrow (R^{-1}Q^{-1})A = I \text{ (Thm 2(a))}$$

By Problem 25 from Exercise 2.3, A is invertible and $A^{-1} = R^{-1}Q^{-1}$.

By Theorem 5, for each b in $\mathbb{R}^n$, the equation $Ax = b$ has a unique solution. $\square$

19) Let A be a lower triangular $n \times n$ matrix with nonzero entries on the diagonal. Show that A is invertible and A^{-1} is lower triangular.

Solution: By Defn. of lower triangular matrix, all entries of A above the main diagonal are zero. Given, each main diagonal entries are non-zero.

~~Therefore~~ If $a_{ii} = \lambda$, then a_{ji} where $j > i$ is changed by Row-
Replacement as follows: $a_{ji} = a_{ji} - a_{ji} \cdot a_{ii}^{-1}$, where $a_{ii}^{-1} = \frac{1}{\lambda} a_{ii}$.

The i th row's main diagonal entry is first scaled down to 1 by first multiplying row i by $\frac{1}{a_{ii}}$. $\therefore$ New $a_{ji} = 0 \forall j > i$.